

Project Design Phase-II

Technology Stack (Architecture & Stack)

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Frontend for users and admin	React.js, HTML, CSS, JavaScript
2	Application Logic-1	User Authentication & Authorization	Node.js, Express.js
3	Application Logic-2	Product & Cart Management	Node.js, Express.js
4	Application Logic-3	Order & Admin Management	Node.js, Express.js
5	Database	Store users, products, orders	MongoDB
6	Cloud Database	Database hosting service	MongoDB Atlas
7	File Storage	Product images	Local Storage / Cloudinary
8	External API-1	Payment Integration	Razorpay API
9	External API-2	Email Notifications	Nodemailer
10	Machine Learning Model Not Applicable		N/A
11	Infrastructure (Server/Cloud)	Application Deployment	Localhost / Render / Vercel

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in project	React.js, Node.js, Express.js
2	Security Implementations	Authentication & data protection	JWT Authentication, Password Hashing
3	Scalable Architecture	Supports future growth	MERN Stack Architecture
4	Availability	Accessible through web browser	React + Node.js Deployment
5	Performance	Fast loading and API responses	MongoDB Indexing, React Optimization

Architecture Diagram

